

ECE160 Final Project Proposal

Aidan Lam & Lucas Teichmann

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1 Project Overview

This project will build a 2D gravitational orbit simulator that models the motion of multiple bodies using realistic gravitational mathematical models, allowing for real-time modification of the simulator.

2 Functional Requirements

Minimal Requirements: Gravitational Orbit Simulator

1. Be able to place customizable bodies¹ (e.g. names, colors, proportions)
2. Be able to alter properties of the bodies mid-simulation (e.g. mass, size, and density)
3. Must be able to handle more than two bodies
4. Show path and trajectory of the bodies
5. Shows changes in real time (changes to bodies, etc.)

Stretch Goals: Rocket Implementation

1. User-piloted rocket in real time with keyboard control
2. Alterable rocket features (fuel efficiency, thrust capacity, size, color, etc.)
3. Shows rocket interaction with bodies in real time
4. Collision handling (rocket-body contact, body-body contact, etc)

¹Bodies refers to celestial objects, such as planets, stars, black holes, etc

3 Implementation

1. C aspects:
 - (a) Numerical differential equation solver (for calculating position and velocity of bodies)
 - (b) Structs Usage: Individual body characteristics (position, velocity, acceleration, mass, size)
 - (c) Physics Calculations (e.g. Gravitational Laws)
2. Python Aspects
 - (a) Graphics/Visualization of simulation (drawing bodies, showing motion, displaying trajectories)
 - (b) User interaction (Creation of bodies, sliders to alter body properties)
 - (c) Interface between C and Python (Collecting inputs for C functions, displaying outputs from C functions)
 - (d) UI Elements (Labels for planets/bodies, buttons/menus for changing settings, display values for mass, velocity, etc.)
 - (e) Libraries (Currently Considering): Pygame, Matplotlib, Tkinter